
SpectraLoRA: Post-Training Alignment for Molecular Foundation Models, Unifying Standardized Datasets and Supporting Peptides

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Abstract

1 Background

It is customary at this stage to develop a comprehensive grasp of the literature before introducing one’s own model. In the spirit of this collegial tradition, we shall attempt a survey of the field, commencing inevitably with Density Functional Theory (DFT).

Though I had success in my research both when I was mad and when I was not, eventually I felt that my work would be better respected if I thought and acted like a “normal” person.
– John Forbes Nash, Jr.

1.1 Foundational Knowledge Curiously Absent from the Middle School Curriculum

One is often told that early education is the crucible in which vital life skills are forged; namely group work, rudimentary algebra, and the avoidance of eye contact. In retrospect, these lessons seem woefully trivial compared to the practical utility of good, old-fashioned quantum chemistry.

It is a profound failure of the modern educational apparatus that, somewhere between lunch and World Cultures, a rigorous seminar on DFT was not instituted as a defining cornerstone of youth. Consequently, for the benefit of those readers whose formative years were similarly deprived of the wonders of Hellman, Feynman, Born, and Oppenheimer, we must now take a brief detour to lay the foundations.

As one should have recalled, the exact determination of the electronic density of molecules would provide access to accurate bonded and non-bonded interatomic interactions by way of the Hellman-Feynman Theorem (Khabibrakhmanov et al., 2025). Density Functional Approximations simply provide approximate electron densities, and therefore the electronic structure, of atomistic systems (Khabibrakhmanov et al., 2025). But we restate the inherently obvious definitions from physics upon which everything must rest.

First, let us quickly build up to the Schrödinger equation. We will state the equation not derive it. An Excellent derivations is already available in (Morin, 2008). Now we proceed with defining the most elementary of definitions.

Definition 1 (Observable). *An observable is any dynamical variable that can be measured.*

Definition 2 (Operator). *An operator is a symbol for an instruction to carry out some action, namely an operation, on a function.*

Note that operators are linear if they satisfy additivity and homogeneity namely, $f(av+bw) = af(v)+bf(w)$. If V is a finite dimensional vector space $\text{Hom}(V)$ is the space of linear operators on V such that $V \otimes V^* \cong \text{Hom}(V)$. Alternatively one may think of a specific field k , and then view k -linear maps between two k -vector spaces as equivalent to k -module homomorphisms. In the context of Quantum mechanics some

more succinctly state that linear operators satisfy the principle of superposition acting on a sum of states $f(c_1\psi_1 + c_2\psi_2) = c_1f(\psi_1) + c_2f(\psi_2)$. All quantum-mechanical operators that represent dynamical variables are hermitian.

Definition 3 (Eigenfunction). *A function f is an eigenfunction of an operator Ω if it satisfies $\Omega f = \omega f$ for constant ω . Recall that a linear combination of degenerate eigenfunction is also an eigenfunction.*

A common property of operators is that the outcome of successive operations depends on the order in which the operations are carried out. In general operations do not commute.

Definition 4 (Commutator). *Let A and B be operators. Then the commutator $[A, B] = AB - BA$.*

From these definitions we can see that observables of interest can be constructed from the operators for position and momentum (Atkins & Friedman, 2011). Namely the kinetic energy operator T comes from noticing the relation between kinetic energy and linear momentum. We write simply,

$$T = \frac{p_x^2}{2m} = \frac{1}{2m} \left(\frac{\hbar}{i} \frac{d}{dx} \right)^2 = -\frac{\hbar}{2m} \frac{d^2}{dx^2} \quad (1)$$

which in higher dimensions is:

$$T = -\frac{\hbar^2}{2m} \nabla^2 \quad (2)$$

The operator for potential energy of a particle in one dimension, typically denoted $V(x)$, is multiplication by the function in the position representation (Atkins & Friedman, 2011). As is succinctly explained by Morin (2008) the position representation the operator for the Coulomb potential energy of an electron e in the field is

$$V = -\frac{Ze^2}{4\pi\epsilon_0 r} \times \quad (3)$$

for nucleus of atomic number Z , and distance from nucleus to electron r . This immediately leads to the Hamiltonian

$$H = T + V \quad (4)$$

which is the operator for the total energy of a system.

Definition 5 (Bracket Notation). *We write integrals as*

$$\langle m|\Omega|n\rangle = \int f_m^* \Omega f_n d\tau \quad (5)$$

where $|n\rangle$ is termed a ket and denotes the state described by the function f_n . The bra is denoted by $\langle m|$ and denotes the complex conjugate of the function f_m^* . Putting together a bra and ket and placing an operator in the middle is termed a bracket, namely $\langle m|\Omega|n\rangle$, and understood to equate to $\int f_m^* \Omega f_n d\tau$.

Proposition 1 (Postulate 1 in (Atkins & Friedman, 2011)). *We classically say that the state of a system is described by a function $\Psi(r_1, \dots, r_n, t)$ where $r_1, \dots, r_n$ are the spatial coordinates of particles $1, \dots, n$ that constitute the system at time t . The function Ψ is called the wavefunction of the system. When we wish to denote time-independence it is common to write it as $\psi(r_1, \dots, r_n)$.*

Moreover, note that the wavefunction of a system will be specified by a set of labels called quantum number (Atkins & Friedman, 2011; Morin, 2008; Szabo & Ostlund, 2012).

Proposition 2 (Postulate 2 in (Atkins & Friedman, 2011)). *Observables are represented by Hermitian operators chosen to satisfy the commutation relations*

$$[q, p_{q'}] = i\hbar\delta_{qq'}, [q, q'] = 0, [p_q, p_{q'}] = 0 \quad (6)$$

where q, q' both denote one of the coordinates x, y, z and $p_q, p_{q'}$ are the corresponding linear momenta.

Proposition 3 (Postulate 3 in (Atkins & Friedman, 2011)). *When a system is described by a wavefunction Ω , the mean value of the observable Ω in a series of measurements is equal to the expectation value of the corresponding operator. The expectation value of an operator Ω for an arbitrary state ψ is denoted*

$$\langle \Omega \rangle = \frac{\int \psi^* \Omega \psi d\tau}{\int \psi^* \psi d\tau} \quad (7)$$

Proposition 4 (Postulate 4 in (Atkins & Friedman, 2011)). *The Born interpretation. The probability that a particle will be found in the volume element $d\tau$ at the point r is proportional to $|\psi(r)|^2 d\tau$.*

Proposition 5 (Postulate 5 in (Atkins & Friedman, 2011)). *As dear old Erwin remarked, the wavefunction $\Psi(r_1, r_2, \dots, t)$ evolves in time according to the equation*

$$i\hbar \frac{d\Psi}{dt} = H\Psi \quad (8)$$

In memory of Lord Rayleigh, recall that we suppose a system described by Hamiltonian H has lowest eigenvalue E_0 thereby leading to the Rayleigh Ratio denoted by $\mathcal{E}$ (Atkins & Friedman, 2011).

$$\mathcal{E} = \frac{\int \psi_{\text{trial}}^* H \psi_{\text{trial}} d\tau}{\int \psi_{\text{trial}}^* \psi_{\text{trial}} d\tau} \quad (9)$$

Theorem 1.1 (Variation Theorem (Atkins & Friedman, 2011)). *For any ψ_{trial} , $\mathcal{E} \geq E_0$ provided the trial function is identical to the true ground-state wavefunction of the system. Then as the world intended, depending of course on ones epistemological beliefs,*

Proof. Let $\psi_{\text{trial}} = \sum_n c_n \psi_n$, where $H\psi_n = E_n \psi_n$.

$$I = \int \psi_{\text{trial}}^* (H - E_0) \psi_{\text{trial}} d\tau \quad (10)$$

$$= \sum_{n,n'} c_n^* c_{n'} \int \psi_n^* (H - E_0) \psi_{n'} d\tau \quad (11)$$

$$= \sum_{n,n'} c_n^* c_{n'} (E_{n'} - E_0) \int \psi_n^* \psi_{n'} d\tau \quad (12)$$

$$= \sum_n c_n^* c_n (E_n - E_0) \geq 0. \quad (13)$$

Note that the final inequality follows from $E_n \geq E_0$ and $|c_n|^2 \geq 0$. It follows that

$$\int \psi_{\text{trial}}^* (H - E_0) \psi_{\text{trial}} d\tau \geq 0,$$

which implies $\mathcal{E} \geq E_0$. □

Theorem 1.2 (Hellmann-Feynman theorem (Atkins & Friedman, 2011)). *Consider a system characterized by a Hamiltonian that depends on a parameter P . The theorem simply states*

$$\frac{dE}{dP} = \left\langle \frac{\partial H}{\partial P} \right\rangle.$$

We suppose that the wavefunction is normalized to 1 for all values of P , in which case

$$E(P) = \int \psi(P)^* H(P) \psi(P) d\tau$$

The derivative of E with respect to P is

$$\frac{dE}{dP} = \int \left(\frac{\partial \psi}{\partial P} \right)^* H \psi d\tau + \int \psi^* \left(\frac{\partial H}{\partial P} \right) \psi d\tau + \int \psi^* H \left(\frac{\partial \psi}{\partial P} \right) d\tau \quad (14)$$

$$= E \int \left(\frac{\partial \psi}{\partial P} \right)^* \psi d\tau + \int \psi^* \left(\frac{\partial H}{\partial P} \right) \psi d\tau + E \int \psi^* \left(\frac{\partial \psi}{\partial P} \right) d\tau \quad (15)$$

$$= E \frac{d}{dP} \int \psi^* \psi d\tau + \int \psi^* \left(\frac{\partial H}{\partial P} \right) \psi d\tau \quad (16)$$

In the last term of the second line, we have employed the hermiticity of H to let it operate on the function standing to its left. The first term on the right of the last line is zero because the integral is equal to 1 for all values of P . The second term is the expectation value of the first-derivative of the hamiltonian. Hence,

$$\frac{dE}{dP} = \left\langle \frac{\partial H}{\partial P} \right\rangle.$$

1.2 Density Functional Theory

1.2.1 The Molecular Problem

The Schrödinger equation describes the quantum-mechanical behaviour of particles, making it the most fundamental equation in chemistry (Scherbela et al., 2022; Szabo & Ostlund, 2012). Quantum Chemsitry is typically formulated around the non-relativistic Schrodinger Equation within the Born-Oppenheimer approximation. Formally, we write the non-relativistic Hamiltonian as

$$H = T_n + T_e + V(\mathbf{r}, \mathbf{R}) \quad (17)$$

Following the derivation from Bircher et al. (2017); Szabo & Ostlund (2012) we can arrive that the BO approximation. We know that the behavior of a closed non-relativistic quantum sysmte is completely characterized by the time dependent schrodinger equatio (TDSE).

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle \quad (18)$$

Where $|\Psi\rangle$ is a state vector and $\hat{H}$ is the Hamiltonian. Traditionally, for a molecular system the time-independent Hamiltonian $\hat{H}_{mol} = \sum_{\alpha} \frac{\hat{\mathbf{p}}_{\alpha}^2}{2M_{\alpha}} + \hat{H}_e$ where $\hat{H}_e$ is the electronic Hamiltonian. $\hat{H}_e = \hat{T}_e + \hat{V}_{ee} + \hat{V}_{eN} + \hat{V}_{NN} = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2} + \sum_{i<j} \frac{1}{\hat{r}_{ij}} - \sum_{\alpha,i} \frac{Z_{\alpha}}{|\mathbf{R}_{\gamma} - \mathbf{r}_i|} + \sum_{\alpha<\beta} \frac{Z_{\alpha}Z_{\beta}}{\hat{R}_{\alpha\beta}}$.

The eigenvalues of the position operators $\hat{\mathbf{r}}$ and $\hat{\mathbf{R}}$ are the electronic and nuclear coordinates described by collective variables. Note that $\hat{\mathbf{p}}$ and $\hat{\mathbf{P}}$ are the corresponding momentum operators. Interactions between electrons and nuclei $\hat{V}_{eN}$, nuclear-nuclear $\hat{V}_{NN}$, and electron-electron $\hat{V}_{ee}$ are represented in the above equation (Bircher et al., 2017).

Born and Oppenheimer proposed an Ansatz to this problem. Fundamentally a non-separable Hamiltonian may be written as a product of two subsystems in particular namely $\hat{H}_{mol} = \hat{H}_{fast} \otimes \hat{H}_{slow}$. One Hamiltonian is due to the fast motion of the electrons and the other is due to the slower nuclear components.

$$\hat{H}_e(\mathbf{R}) |\Phi_t \mathbf{R}\rangle = E_t(\mathbf{R}) |\Phi, \mathbf{R}\rangle \quad (19)$$

where the eigenfunctions $|\Psi \mathbf{R}\rangle$ depend parametrically on $\mathbf{R}$.

Fundamentally $\Psi(\mathbf{r}, \mathbf{R}, t)$ depends on both the nuclear and electronic coordinates and is untractable for most systems. Therefore Born and Oppenheimer proposed an Ansatz. Given that the non-separable Hamiltonian may be represented as $\hat{H}_{mol} = \hat{H}_{fast} \otimes \hat{H}_{slow}$ where one is due to the fast motion of electrons, the other due to slower nuclear components, one can express the spectrum of $\hat{H}_{fast}$ in terms of electronic eigenstates $|\Phi\rangle$ (Bircher et al., 2017). For any nuclear configuration $\mathbf{R}$ Born and Oppenheimer showed

$$\hat{H}_e(\mathbf{R}) |\Phi \mathbf{R}\rangle = E_t |\Phi \mathbf{R}\rangle \quad (20)$$

where the eigenfunctions depend parametrically on $\mathbf{R}$ via $\hat{V}_{eN}$. By orthogonality one obtains the classic factorized expression

$$\Psi(\mathbf{r}, \mathbf{R}, t) = \langle \mathbf{r}, \mathbf{R} | \Psi(t) \rangle = \sum_{\ell} \langle \mathbf{r} | \Phi_{\ell} \mathbf{R} \rangle \langle \mathbf{R}, \Phi_{\ell}; \mathbf{R} | \Psi(t) \rangle = \sum_{\ell} \Phi_{\ell}(\mathbf{r}; \mathbf{R}) \chi(\mathbf{R}, t) \quad (21)$$

the nuclear components $\chi_{\ell}(\mathbf{R}, t)$ or nuclear wavefunctions are projections of the configuration rational basis onto the Hilbert space vector $|\Psi(t)\rangle$, the $\Phi_{\ell}(\mathbf{r}; \mathbf{R})$ are the electronic wave functions of a system with configuration $\{\mathbf{R}\}$.

While the Born-Oppenheimer approximation successfully decouples nuclear and electronic kinematics, solving the electronic time-independent non-relativistic Schrödinger equation remains computationally intractable for most systems (Bartolotti & Flurchick, 1996; Born & Oppenheimer, 2000). Essentially, we think of a macromolecule as a set of quantum objects described by a wave function dependent on spatial and spin degrees of freedom of the electrons, and those of nuclei (Echenique & Alonso, 2007). However, Nuclei are heavier than electrons (Echenique & Alonso, 2007). Classically, we assume Nuclei move slowly in the field of the electrons, which, for each position of the first, immediately adjust their quantum state (Echenique & Alonso, 2007). This is the idea captured in the Born-Oppenheimer approximation. Given that the separation of the electronic and nuclear motions depends on the large difference between the mass of an electron and the mass of a nucleus, one assumes that the motion of the electrons take place in an environment wherein the nuclei are fixed in a configuration. As discussed in his Nobel Prize lecture, Kohn (1999) discusses the presence of an exponential wall. Namely, when the number of atoms exceeds a critical value which is in the neighborhood of $N = 10$ for a system without symmetries. Even for molecular systems in which relativistic effects are insignificant, wherein the wave function is adiabatically partitioned into electronic and nuclear portions; the time-independent non-relativistic Schrödinger equation remains computationally intractable for most systems (Bartolotti & Flurchick, 1996). Therefore, DFT helps us solve the Schrödinger equation through a procedure bypassing the wave function calculation (Bartolotti & Flurchick, 1996). Essentially, DFT simplifies solving the many-body Schrödinger equation by replacing the complex $3N$ dimensional wave function with the 3D electron density as the fundamental variable (Iqbal et al., 2021; Hasnip et al., 2014). Understanding this requires a bit more explanation.

1.2.2 DFT Explained

The classic Many-Body electron Problem can be simply stated. For fixed positions of the nuclei, the electronic Hamiltonian

$$\hat{H}_e = \hat{T} + \hat{V}_{ext} + \hat{V}_{int} + \hat{E}_{II} \quad (22)$$

wherein $\hat{T}$ is the kinetic energy, $\hat{V}_{ext}$ is the potential acting on electrons due to nuclei, $\hat{V}_{int}$ is the classic many-body electron-electron interaction, and $\hat{E}_{II}$ is the ion-ion interaction. Then applying our knowledge of expectation values, the total energy is

$$E = \langle \hat{H}_e \rangle = \langle \hat{T} \rangle + \langle \hat{V}_{int} \rangle + \int d^3r V_{ext}(\mathbf{r}) n(\mathbf{r}) \quad (23)$$

Now from results of Hohenberg and Kohn ground-state properties of a many-electron system are uniquely determined by its electron density rather than the complex many-body wavefunction. The implication of this being that we have a function of three spatial variables rather than a wavefunction in $3N$ dimensions. More accurately stated,

Theorem 1.3 (Hohenberg-Kohn Theorem 1). *For any system of electrons in an external potential $V_{ext}(\mathbf{r})$, that potential is determined uniquely, except for a constant, by the ground state density $n(\mathbf{r})$ (Koch & Holthausen, 2015).*

Theorem 1.4 (Hohenberg-Kohn Theorem 2). *A universal functional for the energy $E[n]$ of the density $n(\mathbf{r})$ can be defined for all electron systems. The exact ground state energy is the global minimum for a given $V_{ext}(\mathbf{r})$, and the density $n(\mathbf{r})$ which minimizes this functional is the exact ground state density (Koch & Holthausen, 2015).*

Consequently, the functional $E[n]$ alone is sufficient to determine the exact ground state energy and density. Excited states of the electrons must be determined by other means. The Hohenberg-Kohn Theorems establish that the ground-state density $\rho_0(\mathbf{r})$ uniquely determines all ground-state properties, and that the total energy $E[\rho]$ is minimized by ρ_0 . However, they do not specify how to compute the universal functional. The Kohn-Sham (KS) formalism enables one to do so by introducing a fictitious system of non-interacting electrons that by construction reproduce the exact ground-state density of the true interacting system (Kohn & Sham, 1965). More formally, KS formalism maps the original interacting-electron problem to a non-interacting one in which the fictitious system retains the same ground-state density as the real one, whereby the quantum properties of the original system are reflected in the Kohn-Sham system by means of a multiplicative potential known as an exchange correlation potential (Garrick et al., 2020; Blinder & House, 2018).

The widely used-version of the Kohn-Sham approach is to assume one can find an auxiliary system of non-interacting electrons with same density as the true interacting system. We can then follow the derivation adapted from Martin (2004). The Hamiltonian for this system can be written:

$$H_{eff} = -\frac{\hbar^2}{2m_e}\nabla^2 + V_{eff}(\mathbf{r}). \quad (24)$$

wherein

$$n_{eff}(\mathbf{r}) = \sum_{i=1}^N |\psi_i(\mathbf{r})|^2, \quad T_{eff} = -\frac{\hbar^2}{2m} \sum_{i=1}^N \langle \psi_i | \nabla^2 | \psi_i \rangle, \quad (25)$$

The Kohn-Sham energy is the sum

$$\tilde{E} = T_{eff} + \int V_{ext}(\mathbf{r}) n_{eff}(\mathbf{r}) dr + E_{Hartree}[n_{eff}] + E_{xc}[n_{eff}], \quad (26)$$

The ground state can be found by minimizing with respect to the independent-electron wavefunctions $\psi_i(\mathbf{r})$

$$\frac{\partial \tilde{E}[n]}{\partial \psi_i(\mathbf{r})} = \frac{\partial T_{eff}}{\partial \psi_i(\mathbf{r})} + \frac{\partial E_{other}}{\partial n_{eff}(\mathbf{r})} \frac{\partial n_{eff}(\mathbf{r})}{\partial \psi_i(\mathbf{r})} = 0 \quad (27)$$

subject to the orthonormalization constraints

$$\langle \psi_i | \psi_j \rangle = \delta_{i,j} \quad (28)$$

The Lagrange multiplier method constraints leads to the Kohn-Sham Schrodinger-like equations:

$$(H_{eff} - \epsilon_i) \psi_i(\mathbf{r}) = 0 \quad (29)$$

where the ϵ_i are the eigenvalues

$$H_{eff}(\mathbf{r}) = -\frac{\hbar^2}{2m_e}\nabla^2 + V_{eff}(\mathbf{r}). \quad (30)$$

and

$$V_{eff}(\mathbf{r}) = V_{ext}(\mathbf{r}) + \frac{\partial E_{Hart}}{\partial n_{eff}(\mathbf{r})} + \frac{\partial E_{xc}}{\partial n_{eff}(\mathbf{r})} = V_{ext}(\mathbf{r}) + V_{Hart}[n_{eff}] + V_{xc}[n_{eff}]. \quad (31)$$

The total energy can be written

$$\tilde{E} = \sum_{i=1}^N \epsilon_i - \frac{1}{2} \int V_{Hart}(\mathbf{r}) n_{eff}(\mathbf{r}) dr + \left(E_{xc}[n] - \int (V_{xc} n_{eff}(\mathbf{r}) dr) \right) \quad (32)$$

The Exchange-Correlation energy is defined by

$$F_{HK}[n] = T[n] + V_{int}[n] = \langle \hat{T} \rangle + \langle \hat{V}_{int} \rangle \equiv T_{eff}[n] + E_{Hartree}[n] + E_{xc}[n] \quad (33)$$

where

$$E_{Hartree}[n] = \frac{1}{2} \int d^3r d^3r' \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \quad (34)$$

Thus the exchange-correlation functional is *defined* to be

$$E_{xc}[n] = \langle \hat{T} \rangle - T_{eff}[n] + \langle \hat{V}_{int} \rangle - E_{Hartree}[n] \quad (35)$$

Essentially, full quantum-mechanical treatment of a molecule treatment is expensive (Hasnip et al., 2014). However, for the vast majority of simulations the behavior of nuclei are decoupled from that of electrons due to mass differences and they may be treated as classical point-like particles, essentially the Born-Oppenheimer approximation (Hasnip et al., 2014). The low mass of electrons implies that one must solve the many-body Schrödinger equation to understand their behavior. However, doing so is intractable and DFT let’s us sidestep this difficulty by consequence of the Hohenberg-Kohn Theorems and Kohn-Sham approach which is explained in the equations above. If you have made it this far congratulations! You are now as aware of DFT as the average middle school student!

1.3 Raman Spectroscopy

1.3.1 Conceptualization & Principles

Raman scattering is a spectroscopic technique wherein one shines a monochromatic light onto a sample and detects the scattered light (Szymanski, 2012). The majority of the scattered light will pass through the sample without interaction. The result is the detector will receive energy that is the same frequency as the excitation source known as Rayleigh scattering (Szymanski, 2012).

1.3.2 How things actually work (sort of)

As discussed in Dall’Osto & Corni (2022)

1.4 Equivariant Graph Neural Networks

As god himself stated an equivariant morphism is a homomorphism of G-actions. An equivariant function is a homomorphism between two G-sets.

1.5 DetaNet, MACE, NequIP, e3nn & other molecular models

1.6 Mixture of Experts

1.7 A Theory of Experimental Alignment (Sim2Real but for molecules)

2 Distributed System

This section describes the distributed infrastructure used to (i) generate and curate molecular training data, (ii) run distributed GPU training at high utilization, and (iii) serve low-latency inference behind a secure API. The design is intentionally *decoupled* across planes so that slow, failure-prone offline computation (e.g., quantum chemistry, conformer generation) does not interfere with high-throughput training or online serving.

2.1 Technologies Background

The intended audience for this publication may not have a background in computer science or distributed systems. We therefore introduce a small set of foundational concepts from first principles. Readers experienced with distributed systems may skip ahead to the requirements in Subsection 2.2.

Definition 6 (Computer). *A computer is a programmable machine composed of (i) processing elements that execute instructions (e.g., CPUs and accelerators), (ii) memory that stores instructions and data (e.g.,*

RAM), and (iii) input/output (I/O) devices for persistent storage and communication (e.g., disks, SSDs, networks). A CPU executes program instructions and typically contains arithmetic/logic units (ALUs), registers, and control logic.

Definition 7 (Instruction Pipelining). *Pipelining divides instruction execution into stages (e.g., fetch, decode, execute, memory, write-back) so multiple instructions can be in-flight simultaneously. In a synchronous pipeline, the clock period is constrained by the slowest pipeline stage (plus pipeline-register overhead), and practical throughput is reduced by hazards (data dependencies, control/branch effects) that introduce stalls.*

Definition 8 (Super-pipelining). *Super-pipelining increases the number of pipeline stages by splitting existing stages into smaller steps. This can increase peak clock frequency because each stage does less work, but it does not guarantee linear speedups: additional stages add overhead and can amplify penalties from hazards (e.g., branch mispredictions). As a result, deeper pipelines may improve throughput only when the workload and microarchitecture (prediction, caching, scheduling) support it.*

Definition 9 (Translation Lookaside Buffer (TLB)). *The TLB is a small hardware cache that stores recent translations from virtual addresses (used by processes) to physical addresses (used by RAM). It is often implemented as content-addressable memory (CAM) whose search key is the virtual address (typically the virtual page number plus an address-space identifier), returning the corresponding physical frame number on a hit. On a miss, hardware and/or the operating system performs a page-table walk to obtain the translation.*

Definition 10 (CALL (Compiler, Assembler, Linker, Loader)). *To execute a high-level program (e.g., C), source code is translated into machine-executable form through multiple stages. A common conceptual pipeline is: **compiler** (source to lower-level representation, often producing assembly or object code), **assembler** (assembly to object files), **linker** (combines object files and libraries into an executable), and **loader** (the operating system component that maps the executable into memory and starts execution). Toolchains often bundle these steps behind a single driver (e.g., gcc) and may skip explicit assembly output depending on options.*

Definition 11 (Operating System and Kernel). *An operating system (OS) manages hardware resources (CPU time, memory, storage, devices) and provides abstractions for applications (processes, files, sockets). The kernel is the privileged core of the OS that implements system calls, schedules processes/threads, manages virtual memory, and mediates access to devices (often via drivers), ensuring isolation and fair resource sharing.*

Definition 12 (NCCL (AllReduce, Broadcast, AllGather, ReduceScatter)). *NVIDIA NCCL is a library of GPU communication primitives used in distributed training. AllReduce aggregates values across ranks (e.g., summing gradients) and returns the reduced result to every rank. Broadcast copies a buffer from a root rank to all ranks. AllGather concatenates values from all ranks and distributes the concatenated result to every rank. ReduceScatter reduces values across ranks and then splits (scatters) disjoint chunks of the reduced result to each rank.*

Definition 13 (DDP and FSDP). *In Distributed Data Parallel (DDP), each rank holds a full model replica, processes different mini-batches, and synchronizes gradients (typically via AllReduce). Fully Sharded Data Parallel (FSDP) shards parameters, gradients, and optimizer state across ranks to reduce per-GPU memory usage, enabling training of larger models.*

Definition 14 (Reverse Proxy). *A reverse proxy is an intermediary server that terminates client connections (often including TLS), enforces routing/authentication policies, and forwards requests to internal services. It can also load-balance across service replicas and improve resilience and security.*

Definition 15 (Containerization). *Containerization packages an application and its dependencies into an isolated runtime environment (a container) that shares the host OS kernel while isolating processes and resource usage. Containers improve portability and deployment reproducibility.*

Definition 16 (Docker). *Docker is a popular container runtime and tooling ecosystem. On Linux, Docker containers rely on kernel primitives (e.g., namespaces for isolation and cgroups for resource control) rather than hardware virtualization. Containers are commonly designed to be stateless, but stateful workloads are also possible (e.g., via volumes); statelessness is an application design choice, not a requirement.*

Definition 17 (Docker Swarm). *Docker Swarm is Docker’s built-in orchestration mode. A swarm is a set of Docker hosts that cooperate as manager nodes (cluster membership and scheduling) and worker nodes*

- **FR2: Cascading Physical Imputation.** When experimental datasets lack complete 3D conformers or derived properties, the system must generate missing fields via a tiered imputation engine (fast learned potentials first, higher-fidelity quantum chemistry as a fallback) and record provenance for every derived field.
- **FR3: Parameter-Efficient Fine-Tuning of an Equivariant Backbone.** The training engine must adapt a compact pretrained GNN to experimental targets using parameter-efficient modules. (The paper introduces a custom LoRA-style adapter designed for the model; details are provided in the modeling section.)
- **FR4: Online Model Serving.** The system must expose a secure REST API capable of receiving inference requests, retrieving the required molecule context (identifiers + precomputed geometry/features), and returning predictions from fine-tuned weights.

Non-Functional Requirements

- **NFR1: Strict Data Provenance & Lineage.** Every tensor feature must carry machine-readable lineage (e.g., `field_source`, `field_imputed`, `field_confidence`). Derived or imputed values must never silently substitute for experimental ground truth in training targets.
- **NFR2: Idempotent Offline Computation (Fault Tolerance).** The data engineering plane runs on ephemeral compute (e.g., spot instances). Jobs must be restart-safe: worker preemption must not corrupt shared state, and reruns must deterministically converge to the same dataset.
- **NFR3: High GPU Utilization (Throughput).** Training must avoid I/O starvation. Data access patterns should prefer large sequential reads over many small random reads, and avoid cross-process coordination that stalls ranks.
- **NFR4: Fail-Fast Trial Supervision (Compute Efficiency).** A supervisor must detect failed runs (NaNs, divergence, dead processes) and terminate early to prevent wasted GPU-hours; optionally it may also terminate runs that empirically plateau under a defined policy.
- **NFR5: Decoupled Serving & Sub-Second Latency.** The serving plane must scale independently from offline planes and meet sub-second response targets. **Client requests must never query cold object storage.** All required geometries/features must be pre-hydrated into hot storage and cacheable state.

2.3 Architecture Walkthrough

The system is deployed as a decoupled **four-plane architecture** (Figure 1). A molecule flows from offline generation, through distributed training, and into an isolated serving tier. A shared storage layer provides a durable “bedrock” for large artifacts, while online serving uses only hot, pre-hydrated data.

Plane A: Shared Storage Bedrock. A POSIX file system (Amazon FSx for Lustre) acts as the shared backbone for large artifacts (sharded tensors, checkpoints). Because network file systems incur per-operation latency, the pipeline is designed to *avoid many tiny reads/writes* and instead prefers coarse-grained shards and sequential access.

Plane B: Offline Data Engineering (Ephemeral CPU Swarm). Raw datasets (HDF5 / CSV / SDF / PDB, etc.) are ingested by a swarm of CPU workers. Each worker executes deterministic partitioned jobs to: (i) standardize molecular identifiers, (ii) compute or retrieve conformers and intermediate physical properties via the cascading imputation engine, and (iii) materialize training-ready shards.

To guarantee idempotency and resumability on preemptible instances, workers append progress to an **append-only manifest** (`manifest.jsonl`) and write immutable shard outputs (e.g., `.pt` or equivalent) directly to the shared bedrock. A lightweight index (e.g., SQLite) is built to map molecule identifiers to shard locations and metadata.

Plane C: Distributed Training & Meta-Scheduling (Ephemeral GPU Swarm). Once the dataset is materialized, a GPU swarm is provisioned for training. A supervisor (Ray Tune or equivalent) orchestrates trials and enforces fail-fast policies. Each trial launches a distributed PyTorch job (`torchrun`) with NCCL collectives and DDP/FSDP as appropriate.

Throughput design: centralized storage + random small reads can thrash and bottleneck network file systems. Training therefore uses an `IterableDataset`-style reader that streams *pre-sharded, pre-randomized* data in coarse chunks. Ranks compute deterministic partitions of shard streams (e.g., based on rank id and world size) to avoid cross-rank coordination and to keep reads sequential. Rank 0 periodically writes checkpoints back to the bedrock.

Plane D: Online Serving (Isolated Swarm). The online tier is isolated from training and offline compute. Requests traverse a TLS-terminating ingress/reverse proxy and are routed to two logically distinct services:

- **API Service Layer (CPU, replicated N times).** Handles authentication, request validation, and metadata lookups.
- **Model Service Layer (CPU/GPU as needed, replicated M times).** Performs the forward pass and returns predictions.

Hot-data policy (NFR5): Serving never queries cold object storage during a request. Instead, a background hydration job *precomputes and stages* the required geometries/features into hot, low-latency formats (e.g., Parquet on hot storage) and warms an in-memory cache (e.g., Redis) keyed by molecule identifier. The API layer consults hot metadata/state (PostgreSQL/Redis) and forwards only validated, cache-resolved requests to the model layer.

Model weights: Model-service containers load the fine-tuned weights *on container startup* (not on first request) to avoid cold-start spikes. Weights are distributed via the shared bedrock or an image/volume mechanism, but the critical path for a request never includes weight I/O.

2.4 Scaling & Tradeoff Discussion

The infrastructure makes explicit tradeoffs to prioritize research velocity, GPU saturation, and serving reliability under domain-specific constraints. By acknowledging the operational realities of our deployment environment, specifically, a moderate expected serving throughput of ~ 25 queries per second (QPS), we deliberately favor architectural simplicity and isolation over premature over-engineering.

Operational Pragmatism: Docker Swarm vs. Kubernetes. Given the target serving load of ~ 25 QPS, deploying a fully managed Kubernetes cluster (e.g., Amazon EKS) would introduce unwarranted operational complexity, maintenance overhead, and baseline cost. Instead, the serving tier utilizes Docker Swarm directly on EC2 instances. This provides sufficient service replication, internal load balancing, and rapid deployment capabilities with a minimal operational footprint. It allows the research team to maintain the infrastructure autonomously without requiring a dedicated platform engineering organization.

Storage Isolation and Blast Radius Minimization. To prevent a high-throughput training job on the shared POSIX file system (Amazon FSx for Lustre) from saturating I/O and degrading online serving, the storage planes are strictly isolated. Model Service containers do not lazy-load weights from the shared network mount. Instead, fine-tuned weights are eagerly pulled from highly durable Object Storage (e.g., S3) onto local container NVMe volumes during initialization. This guarantees that once a replica is healthy, its inference critical path is immune to shared-storage latency spikes.

Ephemeral Local Indexing over Distributed Locking. During offline data engineering (Plane B), workers must map molecule identifiers to physical shard locations. Rather than writing to a centralized

database over a network mount, which would introduce severe locking contention among hundreds of concurrent workers, each worker generates a local, ephemeral SQLite3 index. This shared-nothing approach guarantees lock-free, high-throughput data materialization while relying on the append-only `manifest.jsonl` to aggregate global state safely.

Asymmetric Serving Replicas ($N \neq M$). The serving tier decouples the API Service Layer ($\times N$ replicas) from the Model Service Layer ($\times M$ replicas). API responsibilities (authentication, routing, metadata lookup, and caching) are I/O-bound and scale efficiently on commodity CPU containers. Conversely, the model forward pass is compute-bound and may require specialized instances. Separating these concerns allows us to scale the lightweight API nodes independently of the heavier inference nodes, optimizing cloud spend.

Lock-Free Streaming vs. Perfect Global Shuffle. True global shuffling across distributed workers can induce inefficient random access patterns over shared storage. We instead pre-randomize datasets during offline sharding and stream coarse shards deterministically per rank. This trades a mathematically perfect per-epoch global shuffle for high-throughput sequential reads and maximal GPU utilization, preserving more than sufficient stochasticity for optimization without the devastating I/O penalty.

3 Methodology

4 Analysis

TODO:FIX Step 1: Analysis and numbers off. statistics notebook refactor. Step 2: adjust narrative to focus also on raman peak matching statistics. Step 3: Present correction vs not corrected for anharmonic. Step 4: Results for recovering better intensities.

4.1 Zero-Shot Evaluation of the Multidimensional Quantum Hessian

Standard *ab initio* evaluation of vibrational spectra via Density Functional Theory (DFT) requires self-consistent field (SCF) cycles to determine the energetic ground state, followed by finite-difference displacements or density functional perturbation theory (DFPT) to compute the dynamical matrix (the mass-weighted Hessian). This scales poorly with system size, representing a major bottleneck for high-throughput materials discovery.

Here, we demonstrate that our foundation model achieves robust, zero-shot generalization of the quantum Hessian across an out-of-distribution (OOD) phase space. Evaluated on a blind, 512-molecule sample from the RamanChemBL dataset, strictly disjoint from the training distribution, the model successfully extrapolates the local harmonic curvature of the Born-Oppenheimer potential energy surface.

As shown in Figure 2, the model places 60.6% of all DFT modes within $\pm 10 \text{ cm}^{-1}$ and 43.7% within $\pm 5 \text{ cm}^{-1}$, achieving a median matched-line error of just 3.56 cm^{-1} and a 90th-percentile error of 8.27 cm^{-1} . Crucially, Two One-Sided Tests (TOST, $p < 0.001$) confirm a strict absence of systematic positional bias. The model does not globally red-shift or blue-shift frequencies; it preserves the isotropic harmonic restoring forces globally, acting as a highly accurate, linearly scaling surrogate for the $O(N^3)$ or $O(N^4)$ DFT frequency calculations.

4.2 The Polarizability Bottleneck and Intensity Fidelity

While the energetic topology (frequencies) generalizes robustly, computing full Raman scattering cross-sections demands accurately predicting the derivative of the macroscopic polarizability tensor with respect to normal mode displacements, $\left(\frac{\partial \alpha}{\partial q_i}\right)$.

Unlike the local restoring forces governing the Hessian, the dynamic electronic response to an external electromagnetic field is intrinsically dependent on non-local electronic structure and wave-function delocalization.

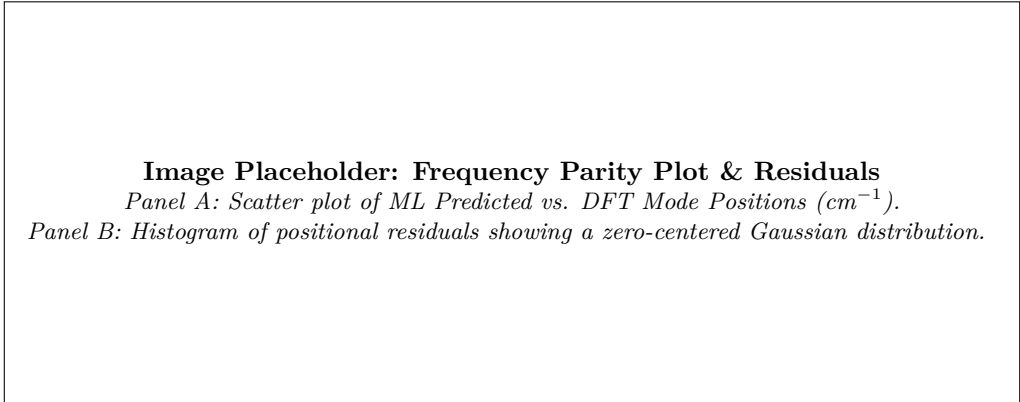


Figure 2: Zero-shot generalization of vibrational frequencies. The model resolves 60.6% of OOD DFT modes within $\pm 10 \text{ cm}^{-1}$, operating fractionally near the intrinsic accuracy limit of the underlying DFT functional.

Consequently, as illustrated in Figure 3, our model, and current foundation architectures broadly, exhibits a fundamental bottleneck in capturing these higher-order derivative properties.

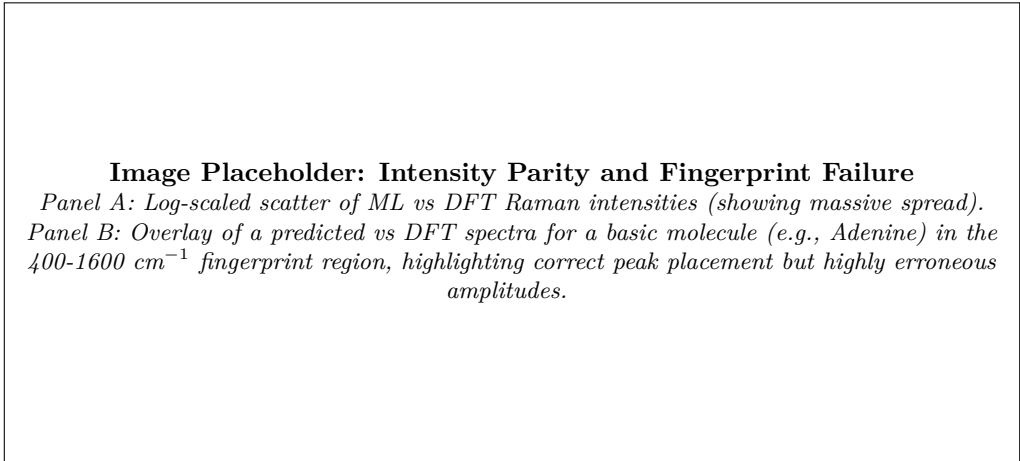


Figure 3: The Polarizability Bottleneck. While normal mode eigenvalues are highly accurate, the predicted intensities (polarizability derivatives) exhibit a log-scale MAE of 0.85, corresponding to an order-of-magnitude physical variance.

The model yields an intensity mean absolute error (MAE) of 0.85 in $\log_{10}$ units against the DFT benchmark, representing a multiplicative error factor of approximately seven. This massive variance in polarizability mapping directly degrades peak-level spectral recovery, particularly in the critical fingerprint region ($400\text{-}1600 \text{ cm}^{-1}$), where intensity-weighted peak coverage drops to 20.3% for DFT and 21.5% against zero-shot experimental data.

We explicitly note that this discrepancy invalidates the use of the raw model outputs for direct experimental Raman fingerprint matching (e.g., distinguishing molecular isomers purely by spectral amplitude profiles). The model successfully learns the geometric harmonic oscillator potentials, but fails to reliably capture the true electron density response required for quantitative scattering cross-sections.

4.3 Utility as a High-Throughput Dynamical Matrix Surrogate

Despite the physical limitations in predicting Raman scattering intensities, the zero-shot accuracy of the predicted Hessian eigenvalues unlocks massive, immediate utility for computational physics and chemistry.

Because thermodynamic corrections, such as Zero-Point Vibrational Energy (ZPE), vibrational entropy (S_{vib}), and thermal free-energy terms, depend exclusively on the vibrational density of states (frequencies) and are entirely agnostic to Raman/IR intensities, this model serves as a direct, ultra-fast replacement for standard DFT in thermochemical pipelines. Furthermore, the sub-4 cm⁻¹ median error allows for rigorous high-throughput transition state validation (identifying imaginary frequencies at saddle points) and massive conformational screening prior to explicit *ab initio* verification.

By decoupling the success of the predicted Hessian from the failure of the predicted polarizability, we establish a new baseline for molecular foundation models: structural and energetic topology is reliably solved OOD, while dynamic response properties remain the critical frontier for future neural architectures.

4.4 Alignment Saves the Day!

4.5 What I would do?

5 Conclusion

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A Appendix

You may include other additional sections here.